COS10005 – Web Development
Lecture 8 – JavaScript

In this session, JavaScript will be introduced.
You may follow the instructions below, to format the HTML and JavaScript with the appropriate elements.
Let's start!

1. Please open the prepared HTML file (Java_Raw.HTML) in the zip folder.
2. Insert an External JavaScript file link under the comment in HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="description" content="Demonstrates for Javascript" />
    <meta name="keywords" content="HTML, Javascript" />
    <meta name="author" content="Your Name"  />
    <!Insert External JavaScript file here>

</head>
```

External JavaScript file link statement: <script src="L8Demo2.js"></script>

3. Now open a new notepad++ and save the file as **"L8Demo2.js"**. The file name has to be same as External JavaScript file link statement: <script src=" **L8Demo2.js**"></script>

JavaScript

4. Simple If statement. A single condition for 2 possible outputs (True or False).

```
function ifTest1() {
var guess;
var secret = 10;
guess = prompt("Enter a number:");
if (guess == secret){
    alert("Correct number: "+secret);
} else {
    alert("Wrong number");
}
}
```

5. To access the output, insert the calling function statement below. First create a function to call another function.

```
function init(){

}
window.onload = init;
```

6. Inside function init(), insert this statement: **ifTest1();** to call function ifTest1() above.

7. Simple If statement. A single condition for 2 possible outputs (True or False). Assign the output to a variable.

```
function ifTest2() {
var guess, msg;
var secret = 10;
guess = prompt("Enter a number:");
if (guess == secret){
    msg = "Correct number: "+secret;
} else {
    msg = "Wrong number";
}
alert(msg);
}
```

8. To access the output, insert: **ifTest2()** inside function init()and remark (**//**) the previous function call.

9. Nested if statement. Multiple if statement to test on 3 conditions for output.

```
function ifTest3() {
var guess, msg;
var secret = 10;
guess = prompt("Enter a number:");
if (guess == secret){
    msg = "Correct number: "+secret;
} else if (guess > secret){
    msg = "Number too high";
} else {
    msg = "Number too low";
}
alert(msg);
}
```

10. To access the output, insert: **ifTest3()** inside function init()and remark (**//**) the previous function call.

11. Nested if statement. Multiple if statement to test on selection condition.

```
function ifTest4() {
var n = prompt("Enter a number");
var ans = "You entered a number between: ";
if (n >= 1 && n <= 9){
      ans = ans + " 1 to 9";
} else if (n >= 10 && n <= 19) {
      ans = ans + " 10 to 19";
} else if (n >= 20 && n <= 29) {
       ans = ans + " 20 to 29";
} else if (n >= 30 && n <= 39) {
        ans = ans + " 30 to 39";
} else if (n >= 40 && n <= 100) {
       ans = ans + " 40 to 100";
} else if (n < 1 || n > 100) {
        ans = "You entered a number less than 1 or greater than 100";
} else {
        ans = "You did not enter a number!";
}
alert(ans);
 }
```

12. To access the output, insert: **ifTest4()** inside function init()and remark (**//**) the previous function call.

13. Similar to nest if statement (sample above), you may also using Switch to work out the process.

```
function switch1() {
 var ans = "";
 var fruitType=prompt("Enter fruit: (Oranges|Apples|Mangoes|Avocadoes)", "Oranges");
switch (fruitType) {
case "Oranges":
        ans = "Oranges are $3.00 a kilo.";
        break;
case "Apples":
        ans = "Apples are $1.99 a kilo.";
        break;
case "Mangoes":
case "Avocadoes":
        ans = "Mangoes and avocadoes are $2.00 each.";
        break;
default:
        ans = "Sorry, we are out of " + fruitType;
}
alert(ans);

 }
```

14. To access the output, insert: **switch1()** inside function init()and remark (**//**) the previous function call.

15. Looping function: While statement. Repeat the process until the condition is met.

```
function while1() {
 var i = 0;
var sum = 0;
while (i < 3){
    i = i + 1;
    sum = sum + i;
    alert ("Value i: " + i + " Sum: "+sum);
}
alert ("Final Value of Sum: "+sum);

 }
```

16. To access the output, insert: **while1()** inside function init()and remark (**//**) the previous function call.

17. Looping function: doWhile statement. Access the body of the loop before (process) before checking on the condition (while statement after the loop).

```
function dowhile1() {
 var i = 0;
var sum = 0;
do {
    sum = sum + i;
    i = i + 1;
    alert ("Value i: " + i + " Sum: "+sum);
} while (i < 3);
alert("Final Value of Sum: "+sum);
}
```

18. To access the output, insert: **dowhile1()** inside function init()and remark (**//**) the previous function call.

19. Looping function with if statement.

```
function dowhile2() {
var guess, msg;
var stop="n";
var secret = 10;
// var secret= Math.floor(Math.random() * (100 - 1)) + 1;
do {
guess = prompt("Enter a number:");
if (guess == secret){
    msg = "Correct number: "+secret;
    stop="y";
} else if (guess > secret){
     msg = "Number too high";
} else {
    msg = "Number too low";
}
alert(msg);
} while (stop=="n");
}
```

20. To access the output, insert: **dowhile2()** inside function init()and remark (**//**) the previous function call.

21. Looping function: For statement set to repeat to the n-times specific in the loop.

```
function for1() {
var i;
var sum = 0;
for (i = 1; i < 3; i++) {
      sum = sum + i;
}
alert(sum);

 }
```

22. To access the output, insert: **for1()** inside function init()and remark (**//**) the previous function call.

23. Looping function: For in statement

```
function forIn() {
var allUnits = ["Creating Web Applications",
 "Web Application Development"];

var oneUnit;
var ans = "";
for (oneUnit in allUnits) {
  ans = ans + " | "+ allUnits[oneUnit];
}
alert(ans);
 }
```

24. To access the output, insert: **forIn()** inside function init()and remark (**//**) the previous function call.

25. Array: Create an index of collection of variables (can be string, numeric, etc).

```
function dispScoreArray(){
    var scores = new Array(3,4,1,5,4);
    var myElement; // array index
    var num; // number
    var ans = ""; // string for output
    var msg = ""; // final message string to be displayed
    // Demonstrates how to use for-in loop
    for (var i=0;i<scores.length;i++){
            num = scores[i];
            ans = i.toString() + ": ";
            for (var j=0; j<num; j++){
                    ans = ans + "*";
            }
            msg = msg + ans + "\n";
    }
    alert(msg);
}
```

26. To access the output, insert: **dispScoreArray()** inside function init()and remark (**//**) the previous function call.